

Abstract

Background: Electronic health records (EHRs) have been shown to support prediction of subsequent antidepressant switching, an imperfect proxy for treatment-resistant depression (TRD). However, it is unclear whether general-purpose narrative embeddings improve prediction beyond structured feature vectors.

Objective: We compared structured feature vectors with pre-trained embeddings of rule-based patient narratives for predicting a treatment-switching proxy for TRD at the index antidepressant prescription.

Methods: This retrospective study included 42,579 patients with depression from one community health system. The outcome required at least 3 distinct antidepressant treatments, including the index agent, within 365 days. Clinical predictors were derived from a 730-day lookback window. We evaluated 4 classifiers on each representation using a shared 80:20 training and test split, with preprocessing and tuning confined to training data. Four embedding encoders were evaluated; Qwen3-Embedding-8B was primary. The pipelines used the same source records. We assessed discrimination, calibration, and paired bootstrap differences, and used concept permutation and neighbor retrieval to examine the predictive signal.

Results: The outcome occurred in 7,455 patients (17.5%, 95% CI 17.2–17.9). In 8,516 test patients, embedded logistic regression achieved a receiver operating characteristic area under the curve (ROC AUC) of 0.657 (95% CI 0.643–0.672), compared with 0.649 (95% CI 0.634–0.664) for feature-vector XGBoost. The post hoc difference between these leading models was 0.008 (95% CI –0.003 to 0.019). Embeddings improved logistic regression by 0.028 (95% CI 0.017–0.039) but reduced discrimination for the 3 tree ensembles, by 0.013 (95% CI 0.001–0.026) to 0.022 (95% CI 0.011–0.033). Across encoders, embedded logistic regression reached 0.645 (95% CI 0.629–0.660) to 0.657. Permuting psychiatric history produced the largest loss for the primary encoder, 0.024 (95% CI 0.010–0.036) to 0.028 (95% CI 0.017–0.039) across classifiers; medication burden and prior treatment exposure also contributed. Permuting race/ethnicity or recorded social determinants changed ROC AUC by no more than 0.003, and every such interval included zero. Nearest-neighbor retrieval stayed below both leading classifiers at every neighborhood size. Importance-weighted retrieval reached 0.625 (95% CI 0.610–0.641) at 295 neighbors, a size chosen on the test patients. Randomly chosen neighbors reached 0.500 (2.5th–97.5th percentile across draws 0.484–0.515).

Conclusions: Narrative embeddings did not demonstrate superior discrimination over the strongest structured feature model for this treatment-switching proxy. Performance depended on the classifier, and neither pipeline established clinical utility. These findings support structured features as a practical benchmark. Before either pipeline is used in care, 3 things need testing: whether the switching label reflects true treatment resistance, whether the models work in other health systems, and whether their predictions improve clinical decisions.

Keywords: treatment-resistant depression; depression; electronic health records; machine learning; clinical prediction; text embeddings; antidepressants